

1.2.2 Richardson Coordinate System for the CRTBP

We introduce here a new coordinate system that proves very useful to compute periodic orbits around Lagrange points (see Section 1.4). As for the Cross coordinate system (see remark 1.2), we define the (Ox) -axis as the line defined by the two primary positions P_1 and P_2 , and the (Oy) -axis as the orthogonal. We choose one Lagrange point L_i , $i \in \llbracket 1, 2 \rrbracket$ as the origin, and we normalize the distance by the distance between the Lagrange point and the second primary P_2 (for the Cross system it is the distance between the two primaries).

Change of Coordinates between Cross and Richardson Systems

We consider one of the 2 collinear points L_i of the CRTBP around the second primary P_2 (L_1 or L_2). We denote by d the distance between the two primaries and by d_i the distance between L_i and P_2 . In this section only, we denote respectively by $(x_C, y_C, z_C, \dot{x}_C, \dot{y}_C, \dot{z}_C)$ and $(x_R, y_R, z_R, \dot{x}_R, \dot{y}_R, \dot{z}_R)$ the coordinates in the Cross and Richardson systems.

To convert a set of coordinates from the Cross system to the Richardson system, the following formula holds

$$\begin{aligned} x_R &= \frac{x_C \pm d_i - (1 - \mu)d}{d_i}, & y_R &= \frac{d}{d_i} y_C, & z_R &= \frac{d}{d_i} z_C, \\ \dot{x}_R &= \frac{d}{d_i} \dot{x}_C, & \dot{y}_R &= \frac{d}{d_i} \dot{y}_C, & \dot{z}_R &= \frac{d}{d_i} \dot{z}_C. \end{aligned}$$

The $+$ sign is for L_1 and the $-$ sign for L_2 .

Conversely, it holds that

$$\begin{aligned} x_C &= \frac{d_i x_R \mp d_i + (1 - \mu)d}{d}, & y_C &= \frac{d_i}{d} y_R, & z_C &= \frac{d_i}{d} z_R, \\ \dot{x}_C &= \frac{d_i}{d} \dot{x}_R, & \dot{y}_C &= \frac{d_i}{d} \dot{y}_R, & \dot{z}_C &= \frac{d_i}{d} \dot{z}_R. \end{aligned}$$

Now that we have introduced the dynamics of the CRTBP, we can study some of its dynamical properties.

1.2.3 Stability by Linearization

Stability properties of the Lagrange points have been discussed by Szebeheley [83]. This is achieved by studying a linearized system. In this section both the planar and the 2D cases are considered.

Planar case

One can easily write the linearization of the dynamics around a Lagrange point L_i , $i \in \llbracket 1, 5 \rrbracket$. We get

$$\begin{pmatrix} \delta \dot{x} \\ \delta \dot{y} \\ \delta \ddot{x} \\ \delta \ddot{y} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{\partial U_\mu}{\partial x \partial x}(\xi_i) & \frac{\partial U_\mu}{\partial x \partial y}(\xi_i) & 0 & 2 \\ \frac{\partial U_\mu}{\partial x \partial y}(\xi_i) & \frac{\partial U_\mu}{\partial y \partial y}(\xi_i) & -2 & 0 \end{pmatrix} \begin{pmatrix} \delta x \\ \delta y \\ \delta \dot{x} \\ \delta \dot{y} \end{pmatrix},$$

where ξ_i is the coordinate of L_i . If we evaluate the matrix of the linearized system at the collinear points L_1 , L_2 and L_3 , we obtain that the matrix has four eigenvalues: two real ones including one which is non negative, and two purely imaginary ones that are conjugates. This proves the instability of the collinear points.

If we evaluate the matrix at the equilateral points L_4 and L_5 , we get that if $\mu < \mu_*$ where

$$\mu_* = \frac{1}{2} \left(1 - \frac{\sqrt{69}}{9} \right) = 0.03851, \quad (1.9)$$

then the matrix has purely imaginary eigenvalues.

If $\mu = \mu_*$, the eigenvalues of the matrix of the linearized system are multiples of $\pm i\sqrt{2}/2$ with non trivial Jordan blocks. Finally, for $\mu > \mu_*$, the eigenvalues are λ , $-\lambda$, $\bar{\lambda}$, and $-\bar{\lambda}$ where λ is not purely imaginary.

In this last case, clearly L_4 and L_5 are unstable. Using results on dynamical systems as in Meyer, Hall, and Offin [62], one can show that L_4 and L_5 are stable when $\mu < \mu_*$. In addition to Szebehely [83], the reader may refer to Bonnard, Faubourg, and Trélat [18] for a precise study of the planar case.

Remark 1.3:

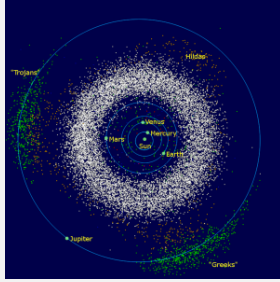


Image Wikipedia.

If we consider the {Sun-Jupiter} system, one can show that the equilateral points are stable and this is coherent with reality.

Indeed, in 1906 a German astronomer Max Wolf discovered the first asteroid captured by the equilibrium region of point L_5 . This kind of asteroid is called a *Trojan asteroid*. In the region of L_4 , it is called *Greek asteroid*. A total of 6,178 Jupiter Trojans had been found in January 2015. The total number of Jupiter Trojans larger than 1 km in diameter is believed to be about 1 million, approximately equal to the number of asteroids larger than 1 km in the asteroid belt.

This illustrates why stable equilibrium points cannot be considered in the design of spacecraft missions because of the collision potential.

3D case

The linearized dynamics around Lagrange points can be written as follows

$$\begin{pmatrix} \delta \dot{x} \\ \delta \dot{y} \\ \delta \dot{z} \\ \delta \ddot{x} \\ \delta \ddot{y} \\ \delta \ddot{z} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ \frac{\partial U_\mu}{\partial x \partial x}(\xi_i) & \frac{\partial U_\mu}{\partial x \partial y}(\xi_i) & \frac{\partial U_\mu}{\partial x \partial z}(\xi_i) & 0 & 2 & 0 \\ \frac{\partial U_\mu}{\partial x \partial y}(\xi_i) & \frac{\partial U_\mu}{\partial y \partial y}(\xi_i) & \frac{\partial U_\mu}{\partial y \partial z}(\xi_i) & -2 & 0 & 0 \\ \frac{\partial U_\mu}{\partial x \partial z}(\xi_i) & \frac{\partial U_\mu}{\partial y \partial z}(\xi_i) & \frac{\partial U_\mu}{\partial z \partial z}(\xi_i) & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \delta x \\ \delta y \\ \delta z \\ \delta \dot{x} \\ \delta \dot{y} \\ \delta \dot{z} \end{pmatrix},$$

where ξ_i is the coordinate of L_i .

- At L_1 , L_2 and L_3 , eigenvalues of this matrix are $(\pm\lambda, \pm i\omega_p, \pm i\omega_v)$, where $\lambda > 0$, $\omega_p > 0$, and $\omega_v > 0$. Since one of these values is non negative, collinear points are unstable.
- For L_4 and L_5 , the spectrum is of the form $(\pm i\omega_1, \pm i\omega_2, \pm i\omega_3)$, where $0 < \omega_1 \leq \omega_2 \leq \omega_3$ and the study of stability is harder and depends on the relation between the values ω_i .

1.3 Jacobi Integral and Hill's Region

In this section, to better convey a physical understanding of the phenomena, we denote the state in terms of position and velocity $(x, y, z, \dot{x}, \dot{y}, \dot{z})$.

1.3.1 Jacobi Integral

The CRTBP dynamics are Hamiltonian and autonomous, hence, an energy integral of motion holds

$$\mathcal{E}(\xi) = \frac{1}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + U_\mu(x, y, z), \quad (1.10)$$

i.e., $\frac{d\mathcal{E}}{dt} = 0$.

The celestial mechanics and dynamical astronomy communities use $-2\mathcal{E}$ which is called the Jacobi integral and is defined from (1.10) as

$$J(x, y, z, \dot{x}, \dot{y}, \dot{z}) = -2\mathcal{E}(\xi) = -(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - 2U_\mu(x, y, z).$$

Note that the time-independency of the Jacobi integral can be derived by a straightforward computation of $\frac{dJ}{dt}$.

1.3.2 Hill's Region

Along trajectories of the CRTBP system, the Jacobi integral is constant, hence one can define the energy surface.

Definition 5 (Energy surface): For a given μ and a constant e , we consider the energy surface, or energy manifold $\mathcal{M}(\mu, e)$, as the five-dimensional surface embedded in the six-dimensional phase space defined by

$$\mathcal{M}(\mu, e) = \{(x, y, z, \dot{x}, \dot{y}, \dot{z}); \mathcal{E}(x, y, z, \dot{x}, \dot{y}, \dot{z}) = e\}. \quad \diamond$$

Then, one can define Hill's region from the previous definition.

Definition 6 (Hill's region): The projection of the energy surface onto the position space in the rotating frame is the region of possible motion for a particle of energy e in the CRTBP gravitational field of mass parameter μ . We denote by $M(\mu, e)$ this projection

$$M(\mu, e) = \{(x, y, z); U_\mu(x, y, z) \leq e\}. \quad (1.11)$$

This is historically known as Hill's region. The boundary of $M(\mu, e)$ is known as the zero velocity curve. $\diamond$

Let us finish by giving one more definition.

Definition 7 (Zero velocity curves): The zero velocity curves (or boundaries of Hill's region) are the locus of the points in the (x, y, z) -space where the kinetic energy vanishes, i.e., for a given energy e and a mass parameter μ

$$\{(x, y, z); e - V_\mu(x, y, z) = 0\},$$

where $V_\mu = \frac{1}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$. $\diamond$

From (1.11), we see that the motion of the spacecraft is possible only on one side of one of these surfaces. Indeed, motion is possible for positive kinetic energy. The other side of the curve is known as the *forbidden realm*.

By multi-variable calculus, one can show that there exist five critical points where the slope of the potential vanishes: three saddle points along the x -axis and two points that are symmetric with respect to the x -axis. Obviously, they correspond to the Lagrange points introduced in Section 1.2.

We denote by $\mathcal{E}_i$ the energy of a particle (a spacecraft) at rest at the libration point L_i ($i \in \llbracket 1, 5 \rrbracket$). It holds that

$$\mathcal{E}_5 = \mathcal{E}_4 > \mathcal{E}_3 > \mathcal{E}_2 > \mathcal{E}_1.$$

Hill's region can be represented as a level section of the graph of the potential U_μ . We plotted the potential for the planar case in Figure 1.4. To have a better understanding of these properties, let us consider the restricted planar case. The extension for the 3D case is easy although we cannot easily represent the different cases.

The intersection of plane $z = e$ with the graph of the effective potential U_μ gives the representation of the different possible zero velocity curves.

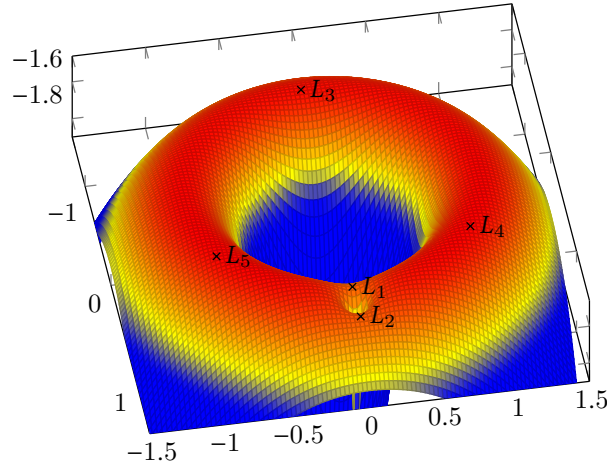


Figure 1.4 – Plot of the potential in the planar case for system {Earth-Moon}. The five equilibrium points L_i , $i \in \llbracket 1, 5 \rrbracket$ are the critical points.

For a given mass parameter μ there are five basic configurations for Hill's region. See Figure 1.5 for a graphical representation.

- Case 1.** $e < \mathcal{E}_1$: If the energy of the spacecraft is lower than $\mathcal{E}_1$, the regions around the two primaries are separated and a spacecraft without control cannot move from one to another. See Figure 1.5a.
- Case 2.** $\mathcal{E}_1 < e < \mathcal{E}_2$: In that case, a “neck” between the realms around the two primaries opens up. We can note that the equilibrium point L_1 is in this neck. The spacecraft is still contained in the *interior realm* and cannot go to the *exterior realm*. See Figure 1.5b.
- Case 3.** $\mathcal{E}_2 < e < \mathcal{E}_3$: Here we have a second opening between realm around the point L_2 . This allows to go from the *interior realm* to the *exterior realm*. See Figure 1.5c.
- Case 4.** $\mathcal{E}_3 < e < \mathcal{E}_4 = \mathcal{E}_5$: Here, there are only two separated forbidden realms around the two equilateral points L_4 and L_5 . A third neck appears around point L_3 . See Figure 1.5d.
- Case 5.** $\mathcal{E}_4 = \mathcal{E}_5 < e$: The forbidden realm disappears. The spacecraft is free to move in the entire (x, y) -plane.

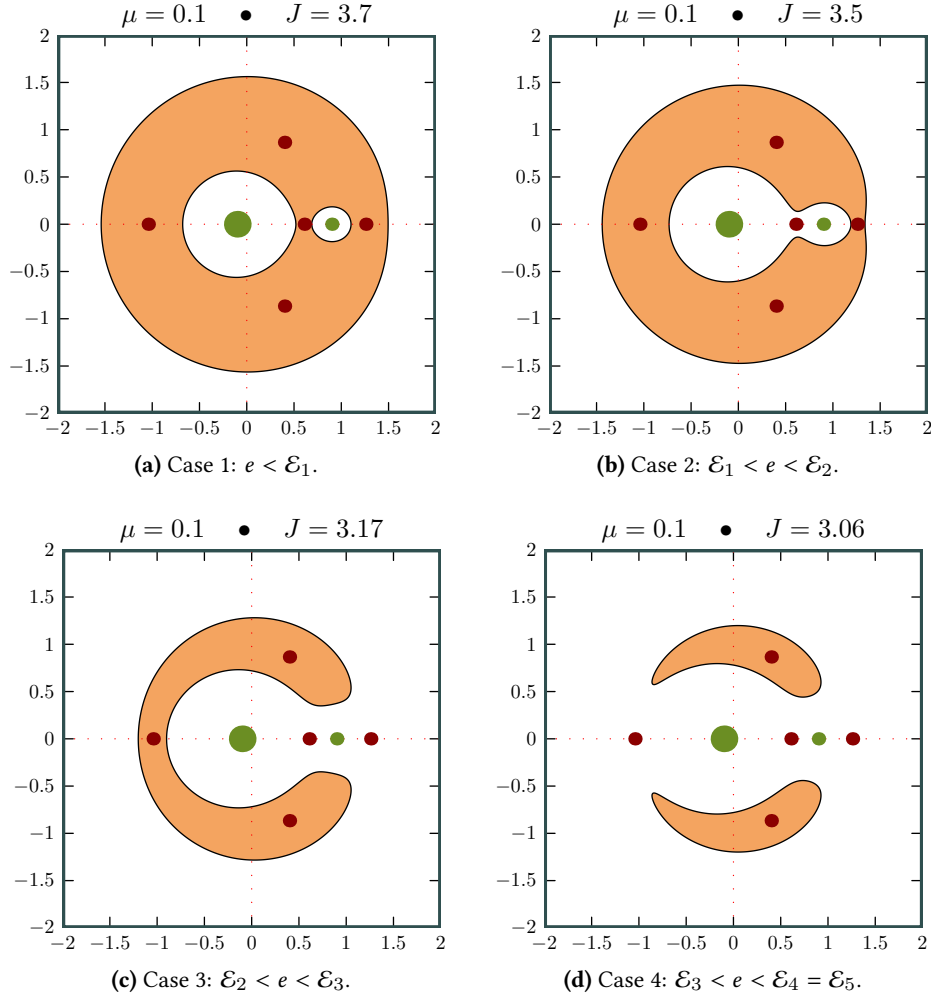


Figure 1.5 – Realms of possible motion (images J.-M. Sarlat).



In this work, we focus on the low thrust transfer in the CRTBP. The study of Hill's regions is crucial because of the good intuition it gives us. Indeed, because we are using low thrust, it will be very interesting to know what regions are accessible without any control. This will become clearer in Section 1.5 about invariant manifolds.

1.4 Periodic Orbits

The equilibrium points in the CRTBP have been known for centuries. As we have seen, Euler in 1767 [34] and Lagrange in 1772 [54] first computed their positions. In 1950, Arthur C. Clarke first suggested that point L_2 could be an ideal place for relaying TV and radio signals.

In 1966, Robert Farquhar discovered trajectories around L_2 on which a communication satellite could be placed to establish a continuous link between Earth and the far side of the Moon. He named such a trajectory a “Halo orbit”. Then NASA became very interested in

the subject and Farquhar published numerous scientific articles covering topics related to Halo orbits (see for instance [35, 36]).

Periodic orbits have been used for real missions such as ISEE-3 (NASA) in 1978, SOHO (ESA-NASA) in 1996, GENESIS (NASA) in 2001, PLANK (ESA) in 2007 etc.

In this section, we will investigate the periodic orbits around the collinear points of the CRTBP. We will introduce the Lyapunov-Poincaré center theorem that ensures that families of periodic orbits exist around the equilibrium points.

1.4.1 Lyapunov-Poincaré Theorem

Let us write the linearized system in the spatial case (the planar case can be easily deduced from it). It is standard to expand the nonlinear terms $\frac{1}{r_1}$ and $\frac{1}{r_2}$ as series in Legendre polynomials, using the formula:

$$\frac{1}{\sqrt{(x-A)^2 + (y-B)^2 + (z-C)^2}} = \frac{1}{D} \sum_{n=0}^{\infty} \left(\frac{\rho}{D}\right)^n P_n\left(\frac{Ax + By + Cz}{D\rho}\right),$$

where $D^2 = A^2 + B^2 + C^2$, $\rho^2 = x^2 + y^2 + z^2$ and P_n is the Legendre polynomial of degree n (see Jorba and Masdemont [50]). Then, using a Richardson coordinate system centered on the equilibrium point L_i introduced in Section 1.2.2, for $i = 1, 2$, the equations of motion (1.6) can be written as

$$\begin{cases} \ddot{x} - 2\dot{y} - (1 + 2c_2)x = \frac{\partial}{\partial x} \sum_{n \geq 3} c_n \rho^n P_n\left(\frac{x}{\rho}\right), \\ \ddot{y} + 2\dot{x} - (c_2 - 1)y = \frac{\partial}{\partial y} \sum_{n \geq 3} c_n \rho^n P_n\left(\frac{x}{\rho}\right), \\ \ddot{z} + c_2 z = \frac{\partial}{\partial z} \sum_{n \geq 3} c_n \rho^n P_n\left(\frac{x}{\rho}\right), \end{cases} \quad (1.12)$$

where

$$c_n = \frac{1}{\gamma_i^{n+1}} \left(\mu + \frac{(1 - \mu)\gamma_i^{n+1}}{(1 - \gamma_i)^{n+1}} \right).$$

As previously, γ_i denotes the distance between the Lagrange point L_i and the second primary.

Hence, at Lagrange points L_i , $i = 1, 2, 3$, the linearized version of (1.12) is

$$\begin{cases} \ddot{x} - 2\dot{y} - (1 + 2c_2)x = 0, \\ \ddot{y} + 2\dot{x} - (c_2 - 1)y = 0, \\ \ddot{z} + c_2 z = 0. \end{cases} \quad (1.13)$$

This is of the kind saddle×center×center. Indeed, a simple computation leads to the eigenvalues $(\pm\lambda, \pm i\omega_p, \pm i\omega_v)$:

$$\lambda^2 = \frac{c_2 - 2 + \sqrt{9c_2^2 - 8c_2}}{2}, \quad \omega_p^2 = \frac{c_2 - 2 - \sqrt{9c_2^2 - 8c_2}}{2}, \quad \text{and} \quad \omega_v^2 = c_2. \quad (1.14)$$

The following theorem ensures the existence of periodic orbits around equilibrium points (see [62, 18] and references therein).

Theorem 1 (Lyapunov-Poincaré center theorem) Let $\dot{x} = H(x)$ be a hamiltonian system in $\mathbb{R}^{2n}$, x_0 an equilibrium point identified with 0, and $A = \frac{\partial H}{\partial x}(0)$ the linearized system matrix. Assume that the eigenvalues of A are

$$\sigma(A) = \{\pm i\omega, \lambda_3, \dots, \lambda_{2n}\},$$

where $\omega > 0$. If $\lambda_j/i\omega \notin \mathbb{Z}$ for $j \in \llbracket 3, 2n \rrbracket$, then there exists a one-parameter family of periodic orbits around 0. Moreover, when approaching the equilibrium point along the family, the periods go to $2\pi/\omega$ and the nontrivial multipliers go to $\exp(2\pi\lambda_j/\omega)$, $j \in \llbracket 3, 2n \rrbracket$.

The multipliers are, by definition, the eigenvalues of the monodromy matrix (see Section 1.5.2), and the non-trivial multipliers are the multipliers left after dropping two $+1$ multipliers from the list of eigenvalues (for references, see [62]).

This theorem can be applied to the planar circular restricted three body problem and to the spatial circular restricted three body problem.

Planar case

Let us apply it to the planar case.

- For the collinear points L_1, L_2 and L_3 , we have seen that the eigenvalues of the linearized system are a pair of real eigenvalues and a pair of pure imaginary eigenvalues (for details, see Meyer, Hall, and Offin [62]). Thus, the Lyapunov-Poincaré theorem implies that there is a one-parameter family of periodic solutions emanating from each of these libration points.
- For the points L_4 and L_5 , for $\mu < \mu_*$ defined in (1.9), we have two pairs of purely imaginary eigenvalues $(\pm i\omega_1, \pm i\omega_2)$ with $0 < \omega_1 < \omega_2$. Applying theorem 1 twice, we obtain the following :
 - since $\omega_2/\omega_1 < 1$: there exists a one-parameter family of periodic orbits emanating from the equilibrium point with a period approaching $2\pi/\omega_1$. This family is called the short period family.
 - If $\omega_2/\omega_1 \notin \mathbb{N}$: there exists a family of periodic orbits emanating from the equilibrium point with period approaching $2\pi/\omega_2$. This family is called the long time period family.

Spatial Case

We have seen that eigenvalues of the spatial circular restricted three body problem are $(\pm\lambda, \pm i\omega_p, \pm i\omega_v)$ with λ, ω_p and ω_v non negative values. Thanks to theorem 1, we obtain the following

- because $\lambda/\omega_v \notin \mathbb{Z}$, there exists a one-parameter family of periodic orbits;
- moreover, if $\omega_p/\omega_v \notin \mathbb{Z}$, we can apply the theorem once again and we obtain another one-parameter family.

When $c_2 > 1$, as for the {Sun-Earth} system, $\omega_p/\omega_v \notin \mathbb{Z}$, and there exists a family of periodic orbits with two parameters around L_1 and L_2 . We will see in the next sections an example of such a periodic orbit: the eight-shape orbit.

Definition 8 (Terminology): *There exists specific terminology for periodic orbits around Lagrange points.*

- In the spatial case, periodic orbits which are homeomorphic to a circle are called *Halo orbits*, other families of periodic orbits are called *Lissajous orbits*.
- In the planar case, periodic orbits which are homeomorphic to a circle are called *Lyapunov orbits*. $\diamond$

1.4.2 Numerical Method to Compute Periodic Orbits

In Richardson coordinates, the dynamics satisfy a property of symmetry: if $\xi(t)$ is a solution of (1.6), then so is $\xi(-t)$. Hence, if there exists a trajectory on one side of the plane $y = 0$, then its mirror part on the other side propagating backward is also a trajectory.

Consider a trajectory beginning at $t = 0$ in the plane $y = 0$ with an orthogonal velocity vector. If there exists a time t_ξ at which the trajectory crosses the plane $y = 0$ with again an orthogonal velocity, then we get a periodic orbit of period $2t_\xi$.

This can be written in the following way: considering a periodic solution of (1.12)

$$\xi(t) = (x(t), y(t), z(t), \dot{x}(t), \dot{y}(t), \dot{z}(t))$$

of period $2t_\xi$, it holds that

$$\xi(0) = (x_0, 0, z_0, 0, \dot{y}(0), 0) \text{ and } \xi(t_\xi) = (x_1, 0, z_1, 0, \dot{y}_1, 0).$$

Shooting Function

To compute these periodic orbits numerically, we use a Newton-like method. More precisely, the problem can be stated as finding the zero of a shooting function. Indeed, let us denote by ϕ the flow of the dynamical system, we search for the initial values for $\xi(0)$ that yield

$$\begin{pmatrix} y(t_\xi) \\ \dot{x}(t_\xi) \\ \dot{z}(t_\xi) \end{pmatrix} = \begin{pmatrix} \phi_2(t_\xi, \xi(0)) \\ \phi_4(t_\xi, \xi(0)) \\ \phi_6(t_\xi, \xi(0)) \end{pmatrix} = \mathbf{0}.$$

Because we have more degrees of freedom than the dimension of the function that we want to nullify, we fix one parameter, usually, the *excursion* z_0 (see Archambeau, Augros, and Trélat [6]). Having done that, the shooting function for this problem can be defined as

$$\mathcal{S}_{z_0}(t_\xi, x_0, \dot{y}(0)) = \begin{pmatrix} \phi_2(t_\xi, \xi(0)) \\ \phi_4(t_\xi, \xi(0)) \\ \phi_6(t_\xi, \xi(0)) \end{pmatrix}. \quad (1.15)$$

To find the solution of $\mathcal{S}_{z_0}(t_\xi, x_0, \dot{y}(0))$, we use a Newton-like method. Of course, to allow such a method to converge, we have to initialize it with a point in the convergence domain. This is the subject of the next section.

1.4.3 Analytical Initialization

To initialize the Newton-like method for finding the zeros of the shooting function defined in (1.15), we initialize the algorithm with an analytical approximation of the periodic orbits that we want to compute.

Lindstedt-Poincaré Method

The method we follow to compute an analytical approximation is the Lindstedt-Poincaré Method (see Poincaré [70] and Lindstedt [57]).

Let us explain the method on the standard example of the Duffing equation

$$\ddot{q} + q + \epsilon q^3 = 0, \quad (1.16)$$

where $\epsilon \ll 1$. For $\epsilon = 0$, we can easily show that a periodic solution is $q(t) = a \cos(t)$ for initial condition $q(0) = a$ and $\dot{q}(0) = 0$.

For $\epsilon \neq 0$, a very naive approach would be to search for solutions of the form

$$q = \sum_{n=0}^{\infty} \epsilon^n q_n(t),$$

with $q_0(0) = a$, $\dot{q}_0(0) = 0$ and $q_n(0) = 0$, $\dot{q}_n(0) = 0$ for every $n \geq 1$. Then, we could solve, recursively, the equations to the same order of ϵ

$$\begin{aligned} \ddot{q}_0 + q_0 &= 0, \\ \ddot{q}_1 + q_1 &= -q_0^3, \\ \ddot{q}_2 + q_2 &= -2q_0^2 q_1, \\ &\dots \end{aligned}$$

Having performed computations, we get for q_1

$$q_1(t) = -\frac{3}{8}a^3 t \sin(t) + \frac{1}{32}a^3(\cos(3t) - \cos(t)).$$

Because of the so-called *secular term* ($t \sin t$), the first-order approximation grows to infinity in time. Hence, this approach is not valid to compute an approximation of the solution.

The Lindstedt-Poincaré method allows for the creation of an approximation that is accurate at all time. A scaled time is defined

$$\tau = \nu(\epsilon)t,$$

where $\nu = 1 + \epsilon \nu_1 + \epsilon^2 \nu_2 + \dots$. Then, with respect to the scaled time, equation (1.16) becomes

$$q'' + \nu^{-2}(q + \epsilon q^3) = 0,$$

with the same initial conditions. Now, a solution is sought in the form

$$q = \sum_{n=0}^{\infty} \epsilon^n q_n(\tau).$$

The obtained zero-th and first order solution in ϵ are

$$q_0(\tau) = a \cos(\tau) \text{ and } q_1(\tau) = (\nu_1 - \frac{3}{8})a^3 t \sin(t) + \frac{1}{32}a^3(\cos(3t) - \cos(t)).$$

The secular term can be removed by making the choice: $\nu_1 = 3/8$. Higher orders of accuracy can be obtained by propagating this perturbation analysis recursively.

In the end of this very short introduction, this method allows for the removal of the secular terms at any approximation order.

Much effort has been put toward computing analytical approximations of Lissajous orbits around Lagrange points using the Lindstedt-Poincaré method. Let us present the remarkable contributions of Jorba and Masdemont [50], Archambeau, Augros, and Trélat [6], and Richardson [74].

Halo Orbit Case

In this section, we recall the method that Richardson used in [74] in 1980. He computed analytical approximations for Halo orbits.

Consider the linearized spatial motion equations (1.13)

$$\begin{cases} \ddot{x} - 2\dot{y} - (1 + 2c_2)x = 0, \\ \ddot{y} + 2\dot{x} - (c_2 - 1)y = 0, \\ \ddot{z} + c_2z = 0. \end{cases}$$

A simple computation yields the eigenvalues of this system: $(\pm\lambda, \pm i\omega_p, \pm i\omega_v)$. If initial conditions are restrained to the non-divergent mode, we obtain solutions for the linearized system of the form

$$\begin{cases} x(t) = -A_x \cos(\omega_p t + \phi), \\ y(t) = \kappa A_x \sin(\omega_p t + \phi), \\ x(t) = A_z \cos(\omega_v t + \psi), \end{cases} \quad (1.17)$$

where $\kappa = \frac{\omega_p^2 + 1 + c_2}{2\omega_p} = \frac{2\lambda}{\lambda^2 + 1 - c_2}$ and, A_x and A_z refer to the excursions respectively in x and z .

One can immediately see that if the ratio between ω_p and ω_v is rational, then the bounded solutions of the linearized system are periodic.

Because there are four purely imaginary eigenvalues, classical theorems stating equivalence between the non-linear system and its linearized form do not hold.

However, a generalization of the Lyapunov theorem, established by Moser [66], ensures that for a certain energy greater than the energy of the considered Lagrange point, the behavior of non-linear solutions is qualitatively equivalent to that of the linear solutions. Hence, periodic orbits in the non-linear system are perturbations of the periodic orbits of the linearized system around Lagrange points.

In equation (1.17), values of the frequencies ω_p and ω_v are fixed and “naturally given” by the parameters of the system and the Lagrange point we consider. The non-linearity can induce changes on these two frequencies. Following the work of Richardson and the Lindstedt-Poincaré method we sketch the different steps needed to obtain an approximation of the Halo orbits for which the ratio between the two proper frequencies is equal to one. For other rational ratios, the orbits are referred to as Lissajous orbits.

The first step is to rewrite the system of equations (1.12), enforcing the relation $\omega_p = \omega_v$ for the linearized part

$$\begin{cases} \ddot{x} - 2\dot{y} - (1 + 2c_2)x = \frac{\partial}{\partial x} \sum_{n \geq 3} c_n \rho^n P_n \left(\frac{x}{\rho} \right), \\ \ddot{y} + 2\dot{x} - (c_2 - 1)y = \frac{\partial}{\partial y} \sum_{n \geq 3} c_n \rho^n P_n \left(\frac{x}{\rho} \right), \\ \ddot{z} + \omega_p^2 z = \frac{\partial}{\partial z} \sum_{n \geq 3} c_n \rho^n P_n \left(\frac{x}{\rho} \right) + \Delta z, \end{cases} \quad (1.18)$$

where $\Delta = \omega_p^2 - \omega_v^2$. Equations (1.18) approximated to the fourth order are

$$\begin{cases} \ddot{x} - 2\dot{y} - (1 + 2c_2)x = \frac{3}{2}c_3(2x^2 - y^2 - z^2) + 2c_4x(2x^2 - 3y^2 - 3z^2) + O(\rho^4), \\ \ddot{y} + 2\dot{x} + (c_2 - 1)y = -3c_3xy - \frac{3}{2}c_4y(4x^2 - y^2 - z^2) + O(\rho^4), \\ \ddot{z} + \omega_p^2 z = -3c_3xz - \frac{3}{2}c_4z(4x^2 - y^2 - z^2) + \Delta z + O(\rho^4). \end{cases}$$

We look for a periodic solution as a series expansion in the excursions

$$\begin{cases} x = Ax_0(\tau) + A^2x_1(\tau) + A^3x_2(\tau) + \dots \\ y = Ay_0(\tau) + A^2y_1(\tau) + A^3y_2(\tau) + \dots \\ z = Az_0(\tau) + A^2z_1(\tau) + A^3z_2(\tau) + \dots \end{cases} \quad (1.19)$$

where $A^n = \sum_{l,p, l+p=n} \lambda_{l,p} A_x^l A_z^p$, and $\lambda_{i,j}$ are the polynomial coefficients. We then define the scaled time $\tau = \nu t$ where

$$\nu = 1 + \sum_{n \geq 1} \nu_n, \quad \nu_n < 1.$$

Recursively, we adjust the parameters ν_n to cancel secular terms that appear in the expansion of each approximation. Equations (1.19) become

$$\begin{cases} \nu^2 x'' - 2\nu y' - (1 + 2c_2)x = \frac{3}{2}c_3(2x^2 - y^2 - z^2) + 2c_4x(2x^2 - 3y^2 - 3z^2) + O(\rho^4), \\ \nu^2 y'' + 2\nu x' + (c_2 - 1)y = -3c_3xy - \frac{3}{2}c_4y(4x^2 - y^2 - z^2) + O(\rho^4), \\ \nu^2 z'' + \omega_p^2 z = -3c_3xz - \frac{3}{2}c_4z(4x^2 - y^2 - z^2) + \Delta z + O(\rho^4). \end{cases}$$

Following the Lindstedt-Poincaré method and with quite long calculations, we obtain that $\nu_1 = 0$ and $\nu = s_1 A_x^2 + s_2 A_z^2$ where s_1 and s_2 are defined in appendix A. We also obtain an approximation of the period of the periodic orbit $T = 2 \frac{\pi}{\omega_p} \nu$.

Unfortunately, all secular terms are not canceled, we have to add two more equations

$$\begin{aligned} l_1 A_x^2 + l_2 A_z^2 + \Delta &= 0 \\ \psi - \phi &= n\pi/2, \quad n \in \{1, 3\} \end{aligned} \quad (1.20)$$

where l_1 and l_2 are also defined in appendix A. For computational details, see [74, 6].

The resulting third order analytical approximation is

$$\begin{cases} x = a_{21}A_x^2 + a_{22}A_z^2 - A_x \cos(\tau_1) + (a_{23}A_x^2 - a_{24}A_z^2) \cos(2\tau_1) + (a_{31}A_x^3 - a_{32}A_x A_z^2) \cos(3\tau_1), \\ y = \kappa A_x \sin(\tau_1) + (b_{21}A_x^2 - b_{22}A_z^2) \sin(2\tau_1) + (b_{31}A_x^3 - b_{32}A_x A_z^2) \sin(3\tau_1), \\ z = \delta_n A_z \cos(\tau_1) + \delta_n d_{21} A_x A_z (\cos(2\tau_1) - 3) + \delta_n (d_{32} A_z A_x^2 - d_{31} A_z^3) \cos(3\tau_1) \end{cases}$$

where $\tau_1 = \omega_p \tau + \phi$, $\delta_n = 2 - n$, n gives the orbit class, which can be interpreted as the direction of rotation. One again, coefficients $a_{i,j}$, $b_{i,j}$ and $d_{i,j}$ are defined in appendix A.

Eight-Shape Periodic Orbit

We will not present the very general method explained in [50, 41] but, an interesting kind of periodic orbit is the “Eight-Shape” periodic orbit. These orbits are nearly vertical and we

will see that their invariant manifolds are very stable compared to the Halo manifolds. This is proved thanks to the study of the local Lyapunov exponents (see Archambeau, Augros, and Trélat [6]).

Despite a mistake in the work in [6]³, the author, following the ideas of Richardson's study, gives an analytical approximation for eight-shape periodic orbits that is accurate enough to initialize a Newton-like method and achieve convergence. The author derives the following expression of the approximation

$$\begin{cases} x = a_{21}A_x^2 + a_{22}A_z^2 - A_x \cos(\tau_1) + (a_{23}A_x^2 - a_{24}A_z^2) \cos(2\tau_1) + (a_{31}A_x^3 - a_{32}A_xA_z^2) \cos(3\tau_1), \\ y = \kappa A_x \sin(\tau_1) + (b_{21}A_x^2 - b_{22}A_z^2) \sin(2\tau_1) + (b_{31}A_x^3 - b_{32}A_xA_z^2) \sin(3\tau_1), \\ z = \delta_r A_z \cos(\tau_2) + \delta_n d_{21} A_x A_z (\cos(2\tau_2) - 3) + \delta_n (d_{32} A_z A_x^2 - d_{31} A_z^3) \cos(3\tau_2), \end{cases}$$

where $\tau_1 = \omega_p \tau + \phi$ and $\tau_2 = \frac{\omega_p}{2} \tau + \psi$. Relations between excursion and phase are the ones introduced in (1.20).

Numerical results

In figures 1.6 and 1.7, we see different periodic orbits for different CRTBPs: the Halo orbit in the {Sun-Earth} system, and an eight-shape orbit in the {Earth-Moon} system. In Figure 1.8, we plot a planar Lyapunov orbit in the {Earth-Moon} system. All examples are computed around L_1 but it could easily be done for L_2 . Thanks to the two analytical approximation for periodic orbits, the Newton-like method converges easily.

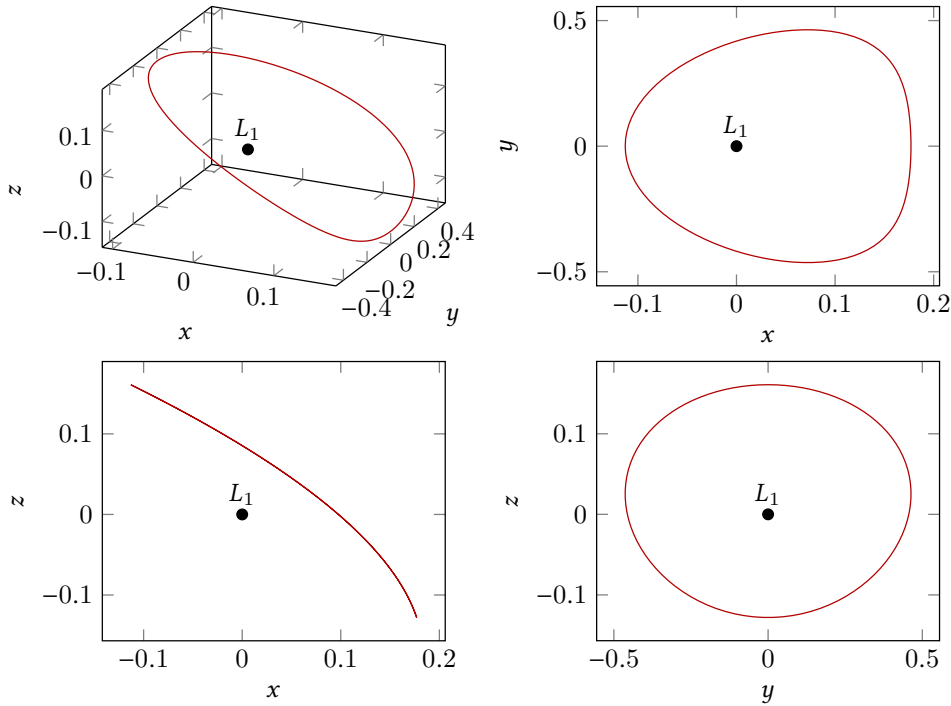


Figure 1.6 – Halo orbit in the {Sun-Earth} system in Richardson coordinates obtained by a Newton-like method. The fixed excursion is $A_z = 240 \times 10^3$ km.

3. Indeed, the argument to cancel the secular term for z_0^2 does not hold for the eight-shape periodic orbit.

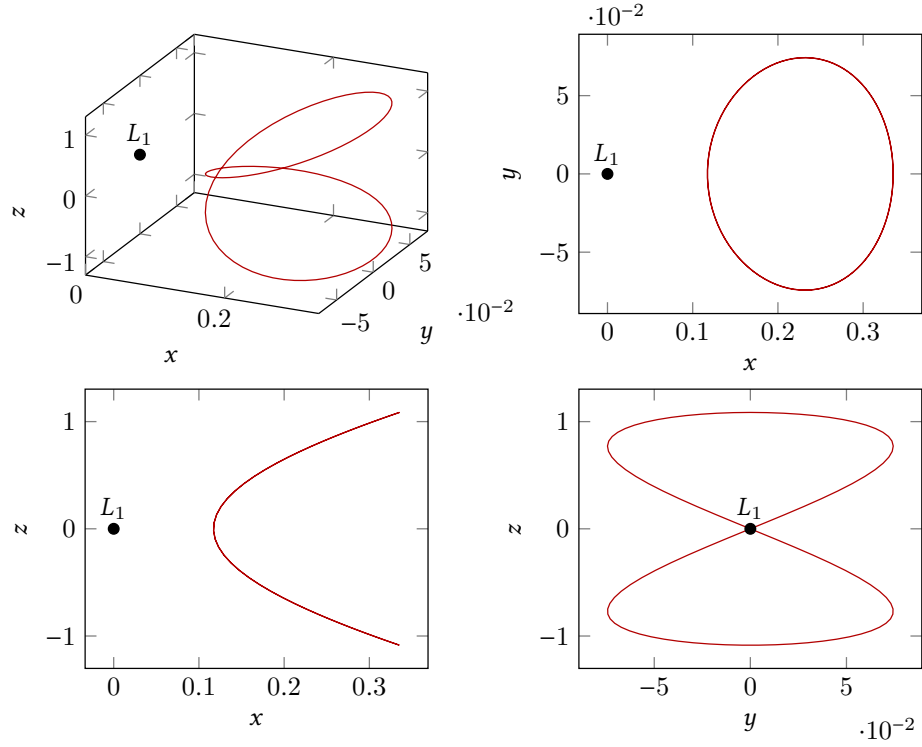


Figure 1.7 – Eight-shape orbit in the {Earth-Moon} system in Richardson coordinates obtained by a Newton-like method. The fixed excursion is $A_z = 630 \times 10^3$ km.

1.4.4 Families of Periodic Orbits

To design spacecraft missions, it is very useful to be able to compute the family of all periodic orbits, providing a variety of orbits that have different energies. In order to achieve this, we first introduce the continuation methods. These methods will be used throughout this thesis, here we introduce the basic concept.

Continuation Method

To compute the family of periodic orbits, we introduce continuation methods also known as homotopy methods. We consider a problem $\mathcal{P}_1$ that we want to solve. The principle is to *embed* this problem into a family of problems depending on a parameter $\lambda \in [0, 1]$: $(\mathcal{P}_\lambda)_{\lambda \in [0, 1]}$. The initial problem $\mathcal{P}_0$ is assumed to be easy to solve, while the final problem $\mathcal{P}_1$ is the one we actually want to solve (and too hard to solve directly).

The continuation method is to follow the zeros of the problems $(\mathcal{P}_\lambda)_{\lambda \in [0, 1]}$ using shooting functions from $\lambda = 0$ to $\lambda = 1$ to finally obtain a solution for $\mathcal{P}_1$. There are many types of continuation methods, for a complete presentation, see Allgower and Georg [3]. First, we consider discrete continuation.

Let us assume that we have solved $\mathcal{P}_0$ numerically, and consider a subdivision $0 = \lambda_0 < \lambda_1 < \dots < \lambda_p = 1$ of the interval $[0, 1]$. The solution of $\mathcal{P}_0$ is used to initialize the Newton-like method applied to $\mathcal{P}_{\lambda_1}$. And so on: solution of $\mathcal{P}_{\lambda_{i-1}}$ is used to initialize $\mathcal{P}_{\lambda_i}$. Of course, the sequence (λ_i) has to be well chosen, and should be refined if necessary. See Section 2.5 for general explanations on discrete continuation methods.

Theoretically, for this method to converge, we need the family of problems to depend

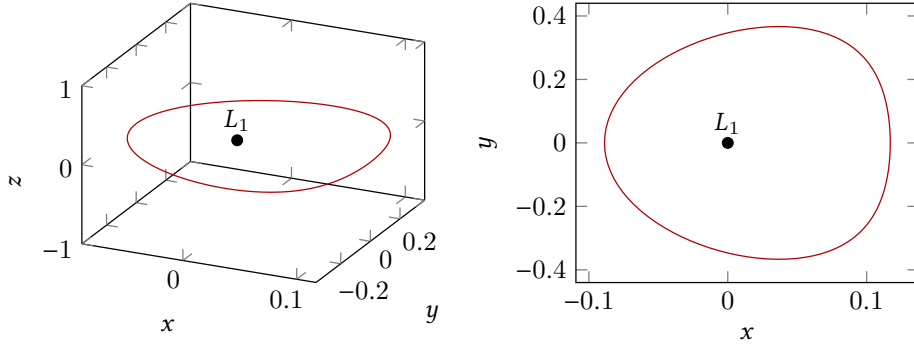


Figure 1.8 – Lyapunov orbit in the {Earth-Moon} system in Richardson coordinates obtained by a Newton-like method. We are in a planar case.

continuously on the parameter λ . See [18, chap. 9] for some justification of the method.

We note that there exist many continuation codes available on the web, such as the well-known Hompack90 [89] or Hampath [30]. For a survey of different results, challenges and issues on continuation methods, see [85].

Application to the family of orbits

Since we had to choose a parameter the excursion z_0 to write the zero function $\mathcal{S}_{z_0}$ in (1.15), it seems natural to use this parameter to perform the continuation that computes the family of orbits. We choose to reach a certain z_0^{obj} (respectively x_0^{obj} in the planar case). So we define our continuation as

$$\mathcal{P}_\lambda : \begin{cases} z_0^\lambda = (1 - \lambda)z_0 + \lambda z_0^{\text{obj}} \\ \mathcal{S}_{z_0}^\lambda(t_\xi, x_0, \dot{y}_0) = \begin{pmatrix} \phi_2(t_\xi, \xi_0^\lambda) \\ \phi_4(t_\xi, \xi_0^\lambda) \\ \phi_6(t_\xi, \xi_0^\lambda) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \end{cases}$$

where $\xi_0^\lambda = (x_0, 0, z_0^\lambda, 0, \dot{y}_0, 0)$. Thanks to the analytical approximation provided by [74] or [50], we can solve the initial problem $\mathcal{P}_0$ as described in the previous section. Note that such an analytical approximation does not work for every z_0 , in particular for small ones. Using the continuation method described previously, we derive a family of periodic orbit.



Unfortunately, for some periodic orbits (Halo family), we observe that the continuation fails when we converge to the equilibrium point ($z_0^{\text{obj}} \rightarrow 0$). A much better continuation parameter is *energy*. It releases the constraint on the parameter z_0 , and allows us to reach any periodic orbit. In particular the algorithm converges to the energy of L_i , $i \in \{1, \dots, 3\}$.

We can explain the failure of the continuation on the z -excursion by the fact that for, energy value close to the one of the libration point, the z -excursion is too small. Moreover, the energy parameter provides monotony to the path of zeros.

Moreover, it is a significantly more natural parameter, keeping in mind the fact that we will construct a controlled transfer method. To the author's knowledge, this is the first time that this continuation is performed with the energy as the continuation parameter. This avoids numerical problems when reaching energies close to L_i .

Thanks to the analytical approximation, we get a first periodic orbit with energy $\mathcal{E}_0$, and

we want to reach a prescribed energy $\mathcal{E}_1$ so we define the following family of problems:

$$\mathcal{P}_\lambda^\mathcal{E} : \mathcal{S}_\mathcal{E}^\lambda(t_\chi, x_0, z_0, \dot{y}_0) = \begin{pmatrix} \phi_2(t_\xi, \xi_0) \\ \phi_4(t_\xi, \xi_0) \\ \phi_6(t_\xi, \xi_0) \\ \mathcal{E}(\xi_0) = \mathcal{E}_\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

where $\mathcal{E}(\xi_0)$ is the energy of the trajectory starting at ξ_0 and

$$\mathcal{E}_\lambda = (1 - \lambda)\mathcal{E}_0 + \lambda\mathcal{E}_1.$$

See Algorithm 1.1 for a precise description of the method.

Algorithm 1.1: Orbit Continuation Algorithm

Require: the objective energy $\mathcal{E}_1$, $\epsilon > 0$, $\epsilon_{\min}$ and $\epsilon_{\max}$.
Require: $\xi_0 = (x_0, 0, z_0, 0, \dot{y}_0, 0)$ and t_ξ provided by the first shoot (with the analytical approximation initialization).
1: $\mathcal{E}_0 = \mathcal{E}(\xi_0)$, $\lambda = 0$.
2: $x_s = x_0$, $z_s = z_0$, $\dot{y}_s = \dot{y}_0$ and $t_s = t_\xi$.
3: $i = 0$.
4: **while** $\lambda < 1$ and $\epsilon > \epsilon_{\min}$ **do**
5: $\epsilon = \min(\epsilon, 1 - \lambda)$
6: we increase : $\lambda \leftarrow \lambda + \epsilon$.
7: we initialize : $(t_c, x_c, z_c, \dot{y}_c) \leftarrow (t_s, x_s, z_s, \dot{y}_s)$
8: $\mathcal{E}_\lambda \leftarrow (1 - \lambda)\mathcal{E}_0 + \lambda\mathcal{E}_1$
9: Newton-like Method to solve $\mathcal{S}_\mathcal{E}^\lambda(t_c, x_c, z_c, \dot{y}_c) = \mathbf{0}$.
10: **if** success **then**
11: we keep the solution : $(t_s, x_s, \dot{y}_s) \leftarrow (t_c, x_c, \dot{y}_c)$
12: we increase : $\epsilon = \min(\alpha\epsilon, \epsilon_{\max})$, $\alpha > 1$.
13: **else**
14: we restore : $\lambda \leftarrow \lambda - \epsilon$.
15: we decrease : $\epsilon = \beta\epsilon$, $\beta < 1$.
16: **end if**
17: **end while**

Numerical Results

In Figures 1.9 and 1.10, we plotted, respectively, the family of Halo orbits for decreasing energies in the {Sun-Earth} system, and the family of eight-shape orbits for decreasing energies in the {Earth-Moon} system. In Figure 1.11, we see the family of planar Lyapunov orbits in the {Earth-Moon} system. All examples are computed around L_1 but the same could easily be done for L_2 . One may note that this continuation is a discrete one with no linear prediction. We can obviously improve this continuation but because the computation time is very small (around 0.01 s including the printing of each orbit during the continuation), we did not add this feature.

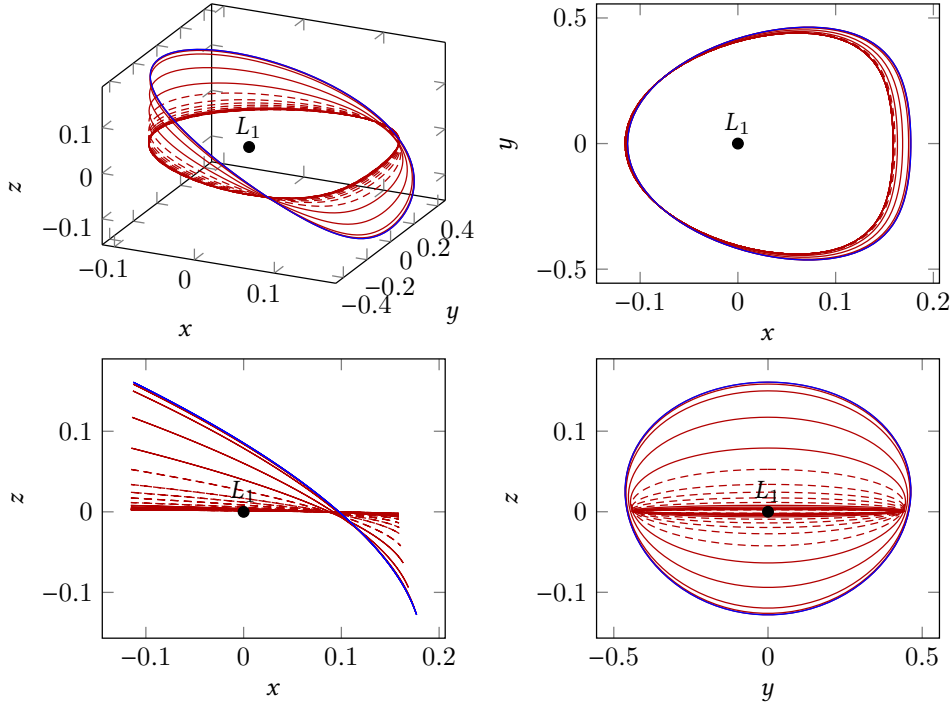


Figure 1.9 – Family of Halo orbits in the {Sun-Earth} system in Richardson coordinates obtained by a Newton-like method and continuation on the energy. The first fixed excursion is $A_z = 240 \times 10^3$ km and the final energy is -1.50042 in normalized units (corresponding to a quasi-null z -excursion).

1.5 Invariant Manifolds

In this section, we introduce the concept of *invariant manifold* of a periodic orbit. In the next chapters, it will become apparent that the existence of such mathematical object is of a great interest. Indeed, the invariant manifolds can be considered as *gravitational currents*, and so, they can be used in interplanetary missions.

1.5.1 Mathematical Definition

We consider a periodic orbit around a libration point L_i , $i \in \llbracket 1, 3 \rrbracket$. Let us begin with the mathematical definition of these objects.

Definition 9 (Invariant Manifolds): *A stable manifold (resp., unstable) of a periodic orbit is the set of the phase space consisting of all points whose future (resp., past) semi-orbits converge to it (asymptotic orbits).* $\diamond$

Because there is no analytical expression for the periodic orbits around libration points, there is no analytical way of computing the associated invariant manifolds. However, the computation of the stable and unstable invariant manifolds can be accomplished numerically. The procedure is based on the monodromy matrix discussed next.

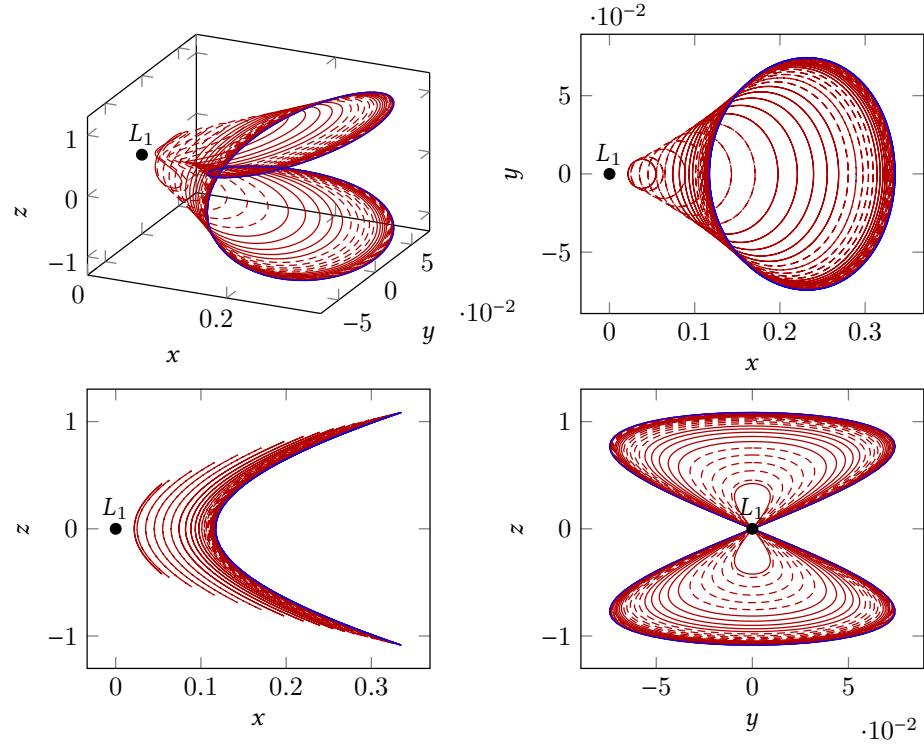


Figure 1.10 – Family of eight-shape orbits in the {Earth-Moon} system in Richardson coordinates obtained by a Newton-like method and continuation on the energy. The first fixed excursion is $A_z = 630 \times 10^3$ km and the final energy is -1.59 in normalized units (corresponding to a quasi-null z -excursion).

1.5.2 Linear Approximation

Let us introduce some basic dynamical tools.

Monodromy Matrix

The monodromy matrix gives us information about the influence of perturbations around the periodic orbit. Let us consider a periodic solution $x^*(\cdot)$ of period T , the flow from the initial point $x_0^* = x^*(0)$ is denoted by $\phi(t, x_0^*)$.

We consider a small perturbation δx_0^* , and we defined the evolution of the perturbation for $t \geq 0$

$$\delta x(t) = \phi(t, x_0^* + \delta x_0^*) - \phi(t, x_0^*).$$

At time T , it holds that

$$\delta x(T) = \phi(T, x_0^* + \delta x_0^*) - \phi(T, x_0^*),$$

which first-order approximation is

$$\delta x(T) = \frac{\partial \phi(T, x_0^*)}{\partial x_0} \delta x_0^*.$$

This leads to the definition of the monodromy matrix.

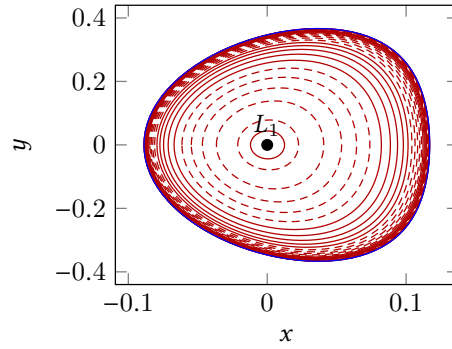


Figure 1.11 – Family of planar Lyapunov orbits in the {Earth-Moon} system in Richardson coordinates obtained by a Newton-like method and continuation on the energy. The final energy is -1.6001 in normalized units.

Definition 10 (Monodromy matrix): *The matrix*

$$M = \frac{\partial \phi(T, x_0^*)}{\partial x_0}$$

determines whether initial perturbations x_0^ from the periodic orbit $x^*(\cdot)$ decay or grow. This matrix is called the monodromy matrix.* $\diamond$

For a complete presentation, see [46].

Next, a well known property of the monodromy matrix is introduced.

Proposition 1 : *Let us consider the dynamical system $\dot{x} = f(x)$ and a periodic solution x^* . The monodromy matrix M is the matrix $\Phi(T)$ where Φ is a solution of the linearized system*

$$\begin{cases} \frac{\partial \Phi}{\partial t} = \frac{\partial f(x^*)}{\partial x} \Phi, \\ \Phi(0) = \text{Id}. \end{cases}$$

Proof : The flow satisfies the following equation

$$\frac{\partial \phi}{\partial t}(t, x_0) = f(\phi(t, x_0))$$

with the initial condition $\phi(0, x_0) = x_0$. We differentiate this equation with respect to the initial condition and obtain

$$\frac{\partial}{\partial t} \frac{\partial \phi(t, x_0)}{\partial x_0} = \frac{\partial f(\phi)}{\partial x} \frac{\partial \phi(t, x_0)}{\partial x_0}.$$

Eigenvalues of Monodromy Matrix

Definition 11 (Floquet multipliers): *The eigenvalues of the monodromy matrix are called the Floquet multipliers (or characteristic multipliers).* $\diamond$

Using the Floquet theorem (see [46]), it can be proved that the monodromy matrix has $+1$ as an eigenvalue with a corresponding eigenvector $f(x_0^*)$ which is tangent to the periodic orbit at x_0^* (see [52, p. 101]).

Drawing a connection between the monodromy matrix and the Poincaré map, one can show that $+1$ is an eigenvalue of the matrix. Moreover, because the CRTBP is a Hamiltonian system (see Meyer, Hall, and Offin [62] and Vogtman, Weinstein, and Arnold [87]), this

eigenvalue has algebraic multiplicity of at least 2. Because the monodromy matrix of an autonomous Hamiltonian system is symplectic, it holds that, if λ is an eigenvalue, then so are λ^{-1} , $\bar{\lambda}$ and $\bar{\lambda}^{-1}$ with the same multiplicity. For details about all these properties, refer to [52, section 4.3].

This leads to the following property for the CRTBP system.

Proposition 2 : *For the CRTBP, eigenvalues of the monodromy matrix satisfy*

$$\lambda_1 > 1, \quad \lambda_2 < 1, \quad \lambda_3 = \lambda_4 = 1, \quad \lambda_5 = \bar{\lambda}_6, \quad |\lambda_5| = 1,$$

where λ_1 and λ_2 are reals.

Note that this is true for the restricted planar system without the complex eigenvalues λ_5 and λ_6 .

Stability of Periodic Orbits

Let us introduce another dynamical tool: the Poincaré map.

Definition 12: *Let Σ be a section surface of co-dimension 1 chosen in such a way that all trajectories that cross Σ in a neighborhood of $q \in \Sigma$ meet two requirements*

- *the trajectories intersect Σ transversely,*
- *they cross Σ in the same direction.*

Let $t_\Sigma(q)$ be the time taken for a trajectory $\phi(t, q)$ to first return to Σ . The Poincaré map is defined by

$$P(q) = \phi(t_\Sigma(q), q). \quad \diamond$$

We remark that a periodic orbit is a fixed point of $P(\cdot)$.

Note that the monodromy matrix has 1 as an eigenvalue with corresponding eigenvectors tangent to the transversal periodic orbit. Since this eigenvector is not in Σ , one can choose an appropriate basis such that the remaining $(n - 1)$ eigenvalues of M are those of $\frac{\partial P(q^*)}{\partial q}$. These eigenvalues are independent of the choice of Σ . As a result of this independence, the periodic orbit is stable if the remaining $(n - 1)$ eigenvalues of M are smaller than 1 in modulus. The orbit $x^*(\cdot)$ is unstable if there exists one eigenvalue λ with $|\lambda| > 1$. The eigenvalues of the Poincaré map determine what happens to small perturbations around q^* within Σ .

1.5.3 Numerical Computation

Local Approximation

Using the Poincaré map it can be shown that the eigenvectors corresponding to eigenvalues other than 1 of the monodromy matrix are linear approximations of the invariant manifolds of the periodic orbit. The eigenvector associated with eigenvalue λ_1 is in the unstable direction and the eigenvector associated with eigenvalue λ_2 is in the stable direction.

Algorithm

Based on that, the method to compute invariant manifolds is the following:

1. First, for ξ_0 a point of the periodic orbit, we compute the monodromy matrix and its eigenvectors. Let's denote by $Y^s(\xi_0)$ the normalized stable eigenvector and by $Y^u(\xi_0)$ the normalized unstable eigenvector.

2. Then, let

$$\begin{aligned}\xi^{s\pm}(\xi_0) &= \xi_0 \pm \alpha Y^s(\xi_0), \\ \xi^{u\pm}(\xi_0) &= \xi_0 \pm \alpha Y^u(\xi_0),\end{aligned}\tag{1.21}$$

be the initial guesses for (respectively) the stable and unstable manifolds. The magnitude of α should be small enough to be within the validity of the linear estimate but not too small to keep a reasonable time of escape or convergence (for instance, see [42] for a discussion on the value of α).

3. Finally, we integrate numerically the unstable vector forward in time, using both α and $-\alpha$ to generate the two branches of the unstable manifold denoted by $W^{u\pm}(\xi_0)$. We do the same for the stable vector only backward in time, and we obtain the two branches of the stable manifold $W^{s\pm}(\xi_0)$ (see Figure 1.12).

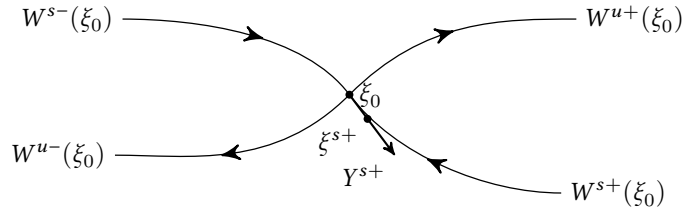


Figure 1.12 – Illustration of the method to compute invariant manifolds.

With this procedure, we are able to compute the invariant manifolds of any Lissajous, Halo, or Lyapunov orbit for any energy greater than the energy of L_i . The complete algorithm is described in Algorithm 1.2. We present an algorithm to propagate the invariant manifolds during a given time parameter. We can easily adjust this algorithm to propagate until a certain plane, for instance the set defined by $x = 1 - \mu$, etc. (see Chapter 2).

Algorithm 1.2: Numerical Computation of Invariant Manifolds

Require: α : parameter defined in (1.21) (depends on the CRTBP)
Require: $(\xi_i)_{i \in I}$: set of points of the considered periodic orbit
Require: T : time of propagation

- 1: **for all** $i \in I$ **do**
- 2: Compute the monodromy matrix by finite differences
- 3: Compute the normalized stable eigenvector $Y^s(\xi_i)$ and the normalized unstable eigenvector $Y^u(\xi_i)$
- 4: Perturb in the four directions to get four branches

$$\begin{aligned}\xi_i^{s\pm} &\leftarrow \xi_i \pm \alpha Y^s(\xi_i), \\ \xi_i^{u\pm} &\leftarrow \xi_i \pm \alpha Y^u(\xi_i)\end{aligned}$$
- 5: Propagate the trajectories with the initial conditions $\xi_i^{s\pm}$ and $\xi_i^{u\pm}$

$$\phi(T, \xi_i^{s-}), \quad \phi(T, \xi_i^{s+}), \quad \phi(T, \xi_i^{u-}), \quad \phi(T, \xi_i^{u+}).$$
- 6: **end for**

Some examples

In appendix B, we present a software developed in FORTRAN90 for Airbus Defence and Space. This software computes periodic orbits with prescribed energy, and its associated invariant manifolds.

Invariant manifolds can be seen as *tubes*, especially in the spatial case. Here we plot some manifolds coming from the periodic orbits that we presented in the previous sections. In Figure 1.13, we plotted the manifolds for the previous Halo orbit with an energy of -1.50042 . We propagate the manifolds during a time equal to 5.0 in the normalized system. The perturbation parameter introduced in (1.21) is $\alpha = 1 \times 10^{-4}$.

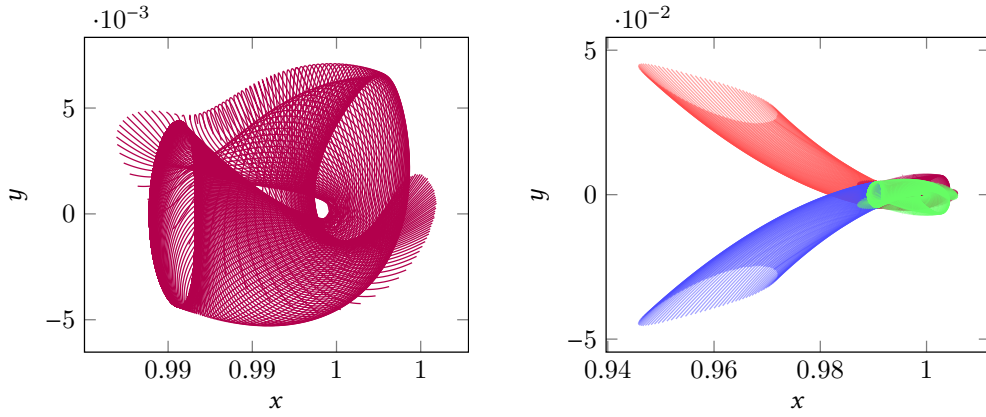


Figure 1.13 – Invariant manifolds of energy $\mathcal{E} = -1.50042$ of the Halo orbit around L_1 in the {Sun-Earth} system. On the left: an unstable manifold near the second primary. On the right: the four manifolds. The propagation time is equal to 5.0 in the normalized system, and the perturbation parameter α is equal to 1×10^{-4} .

In Figure 1.14, we plot the manifolds for an “eight-shape” periodic orbit around L_1 in the {Earth-Moon} system. We propagate the manifolds during a time of 6.0 in the normalized units. We used the same perturbation parameter as for the Halo manifolds.

To finish, we plot in Figure 1.15 the invariant manifolds of a Lyapunov orbit (in planar case) in the {Earth-Moon} system. We have chosen an energy of -1.60 . We propagate the invariant manifolds during a time equal to 8.0. Once again, we used the same perturbation parameter α .

1.5.4 Stability of Manifolds

The study of the stability of invariant manifolds is of great interest. Indeed, in order to design spacecraft missions using invariant manifolds, the general rule is: the more stable the manifolds, the better they can be followed. There are two kinds of stability. The first one is shape conserving stability. We observe in with Figures 1.13, 1.14 and 1.15 that this stability depends on the system considered and the periodic orbit type. We see that manifolds of Halo orbits are not very stable and cannot be used for long time propagation whereas the invariant manifolds of eight-shape orbits are very stable but with low amplitude. Finally, the invariant manifolds of Lyapunov orbits are stable but for very low energy.

Another stability concept is measured by Lyapunov exponents. The concept of Lyapunov exponents (also known as characteristic exponents) was introduced in [59] to investigate the stability properties of solutions of differential equations, and has been used extensively and studied in the literature. Lyapunov exponents measure the exponential

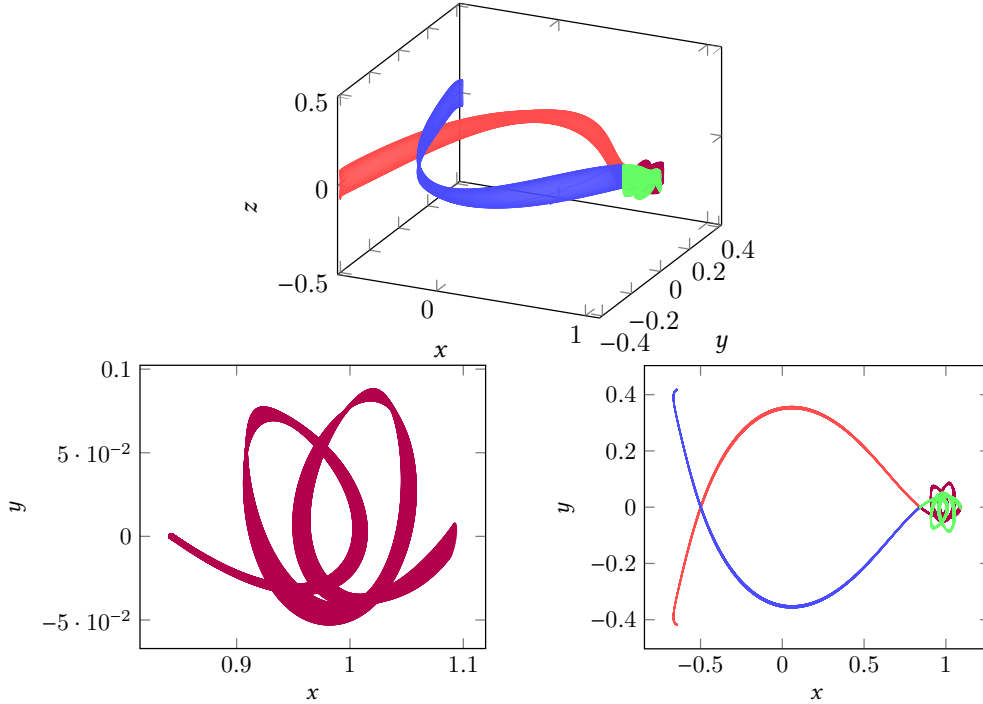


Figure 1.14 – Invariant manifolds of the eight-shape orbit around L_1 in the {Earth-Moon} system of energy $\mathcal{E} = -1.59$. Top: a 3D view of the manifolds. Left: an unstable manifold near the second primary. Right: The four manifolds. The propagation time is equal to 6.0 in normalized system, and the perturbation parameter α is equal to 1×10^{-4} .

convergence or divergence of nearby trajectories in a dynamical system, and provide indications on the behavior in large time of solutions under infinitesimal perturbations. We refer to Archambeau, Augros, and Trélat [6] for a study in the case of the eight-shape orbits.

1.5.5 Theoretical Existence of Invariant Manifolds as Separatrices

To finish this chapter, we sketch the study of the linear flow around equilibrium points. Indeed, this study gives us crucial information for designing spacecraft missions. Studying the linearization of the dynamics near the equilibria is an essential ingredient for understanding the more complete nonlinear dynamics. It can be shown that for a value of energy just above that of L_i (where L_i is the equilibrium point we are considering), the nonlinear dynamics in the equilibrium region is qualitatively the same as the linearized one, we will briefly describe this in the remainder of this section. See [29, 46, 50] and [52, chap. 9].

Canonical Transformation

By a canonical (symplectic) transformation, the linearized Hamiltonian can be written. It is the second-order part of the general Hamiltonian

$$H_2 = \lambda q_1 p_1 + \frac{\omega_p}{2} (q_2^2 + p_2^2) + \frac{\omega_v}{2} (q_3^2 + p_3^2), \quad (1.22)$$

where λ , ω_v and ω_p are the ones defined in (1.14). Variables q_i and p_i are the canonical variables, see Jorba and Masdemont [50] for the details of the canonical transformation.

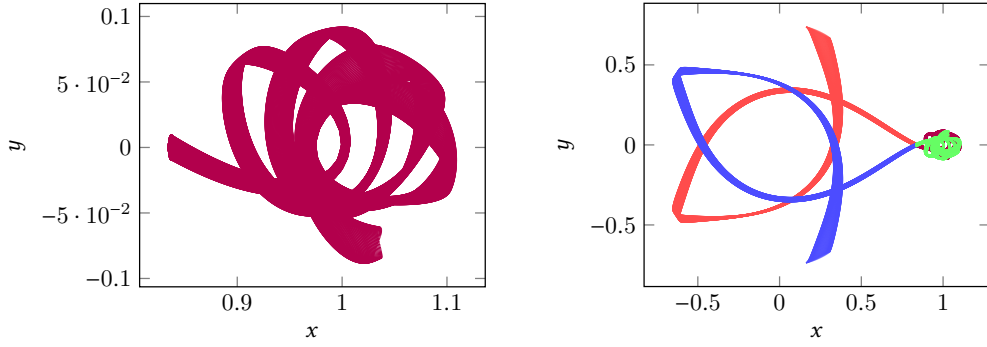


Figure 1.15 – Invariant manifolds of the Lyapunov orbit in the {Earth-Moon} system of energy $\mathcal{E} = -1.60$. Left: an unstable manifold near the second primary. Right: The four manifolds. The propagation time is equal to 8.0 in the normalized system, and the perturbation parameter α is equal to 1×10^{-4} .

A short computation gives the linearized equations in the form

$$\begin{aligned} \dot{q}_1 &= \lambda q_1, & \dot{p}_1 &= -\lambda p_1, \\ \dot{q}_2 &= \omega_p p_2, & \dot{p}_2 &= -\omega_p q_2, \\ \dot{q}_3 &= \omega_v p_3, & \dot{p}_3 &= -\omega_v q_3. \end{aligned}$$

Then solutions can easily be written as

$$\begin{aligned} q_1(t) &= q_1^0 e^{\lambda t}, & p_1(t) &= p_1^0 e^{-\lambda t}, \\ q_2(t) + ip_2(t) &= (q_2^0 + ip_2^0) e^{-i\omega_p t}, \\ q_3(t) + ip_3(t) &= (q_3^0 + ip_3^0) e^{-i\omega_v t}. \end{aligned}$$

Invariant Manifolds as Separatrices

For positive h and c , we defined the region denoted by $\mathcal{R}$ as

$$H_2 = h, \quad |p_1 - q_1| \leq c.$$

This region is homeomorphic to the product of a 4-sphere by an interval I : $\mathbf{S} \times I$. Indeed, for each value of $(p_1 - q_1)$ in $[-c, c]$, the equation (1.22) can be written as

$$\frac{\lambda}{4}(q_1 + p_1)^2 + \frac{\omega_p}{2}(q_2^2 + p_2^2) + \frac{\omega_v}{2}(q_3^2 + p_3^2) = h + \frac{\lambda}{4}(q_1 - p_1)^2.$$

The bounding 4-sphere of $\mathcal{R}$ for $p_1 - q_1 = -c$ is denoted by n_1 and the one for $p_1 - q_1 = c$ by n_2 .

Following the nomenclature of [52], we call the set of points where $q_1 + p_1 = 0$ the *equator* and the sets where $q_1 + p_1 > 0$ and $q_1 + p_1 < 0$, the *northern* and the *southern hemispheres*, respectively.

Classification of orbits for the linear flow. To analyze the flow in $\mathcal{R}$, we consider the different projections onto the (q_1, p_1) -plane, the (q_2, p_2) -plane, and the (q_3, p_3) -plane. In Figure 1.16, we only plot the representation of the *saddle* projection onto the (q_1, p_1) -plane, the two other projections are only circles.

Note that each point in the projection (q_1, p_1) -plane corresponds to a 3-sphere $\mathbf{S}^3$ in $\mathcal{R}$

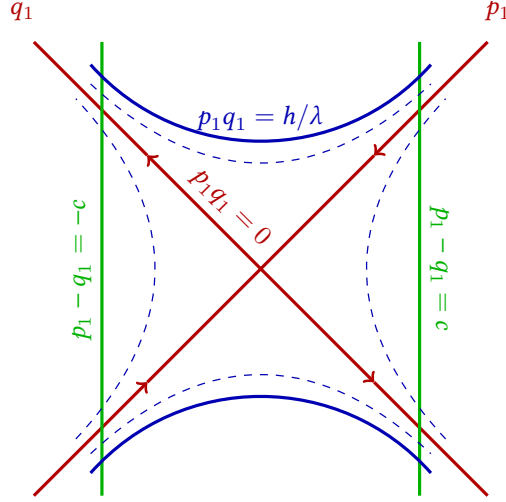


Figure 1.16 – Saddle projection of the flow in the equilibrium region. Only the projection onto the (p_1, q_1) -plane (with axis titled by 45°).

given by

$$\frac{\omega_p}{2}(q_2^2 + p_2^2) + \frac{\omega_v}{2}(q_3^2 + p_3^2) = h - \lambda q_1 p_1.$$

We distinguish 4 categories of orbits:

1. The points $q_1 = p_1 = 0$ correspond to an invariant 3-sphere $\mathbf{S}_h^3$ of *bounded orbits* (periodic and quasi-periodic) in $\mathcal{R}$.
2. The four half open segments on the axes, $q_1 p_1 = 0$, correspond to four cylinders of orbits asymptotic to the previous invariant 3-sphere ($q_1 = p_1 = 0$). They are called *asymptotic orbits* and form the stable (for $p_1 = 0$ and increasing time) and the unstable (for $q_1 = 0$ and decreasing time) manifolds presented in definition 9. The stable manifolds are given by

$$\frac{\omega_p}{2}(q_2^2 + p_2^2) + \frac{\omega_v}{2}(q_3^2 + p_3^2) = h, \quad q_1 = 0.$$

We denote by $W_+^s(\mathbf{S}_h^3)$ the branch going from right to left ($p_1 > 0$) and by $W_-^s(\mathbf{S}_h^3)$ the branch going from left to right ($p_1 < 0$).

The unstable manifolds are given by

$$\frac{\omega_p}{2}(q_2^2 + p_2^2) + \frac{\omega_v}{2}(q_3^2 + p_3^2) = h, \quad p_1 = 0.$$

We denote by $W_+^u(\mathbf{S}_h^3)$ the branch going from right to left ($q_1 > 0$) and by $W_-^u(\mathbf{S}_h^3)$ the branch going from left to right ($q_1 < 0$).

3. The hyperbolic segments defined by $q_1 p_1$ equal to a positive constant corresponds to orbits which cross the region $\mathcal{R}$ from one bounding 4-sphere to the other: $p_1 - q_1 = -c$ to $p_1 - q_1 = +c$ in the southern hemisphere and $p_1 - q_1 = +c$ to $p_1 - q_1 = -c$ in the northern hemisphere. Because these orbits transit from one realm to another, we call them *transit orbits*.

4. Finally, the hyperbolic segments defined by $q_1 p_1$ equal to a negative constant corresponds to orbits connecting the two hemispheres on the same bounding-sphere n_1 or n_2 . Since these orbits return to the same realm, we call them *non-transit orbits*.

Using the *reduction to the center manifold*, one can show that these results can be extended for the nonlinear dynamics. For details, see [52, sec. 9.8].

Remark 1.4: Invariant Manifolds as Separatrices

The main purpose of this section is that, together, the asymptotic orbits form 4-dimensional stable and unstable manifolds, that can be seen as “tubes” $\mathbf{S}^3 \times \mathbf{R}$ to the invariant 3-sphere $\mathbf{S}_h^3$ in a 5-dimensional energy surface. Moreover, as we have seen, these manifolds separate two distinct types of orbits: transit and non-transit orbits. The transit orbits are those *inside* the 4-dimensional manifold tube. The non-transit orbits are those outside the tube.

Let us remark that this is coherent with the plot of the Hill region (see Section 1.5). The forbidden realm can be seen as barrier for non-transit orbits.

This is a very useful property to construct spacecraft missions. Indeed, by computing the manifolds and interpreting them as *gravitational currents*, one can easily find which trajectories realize a transit path or not. This will be the focus of the next chapter.